

1. AutoCure: Concrete Curing ke liye Agentic IoT

The Problem (Masla)

Commercial construction mein mausam ki wajah se delays aur lakho ka nuksan hota hai. Concrete ko mazboot hone (cure hone) ke liye ek khaas temperature aur nami chahiye hoti hai. Abhi mazdoor manually heaters ya paani ka istemal karte hain, jis se aksar errors aate hain aur building ka structure weak reh jata hai.

How It Works (Ye kaam kaise karta hai)

Concrete daalte waqt uske andar saste IoT temperature aur moisture sensors daal diye jate hain. Ek AI agent curing process ko monitor karta hai, local weather forecast check karta hai, aur khud-ba-khud industrial heaters ya water-valves ko control karta hai taake real-time mein perfect temperature maintain rahay.

Why It's Novel (Ye naya kyun hai)

Aam university projects sirf alert dete hain ke "concrete kharab ho raha hai". Ye project ek AI agent use karta hai jo khud action leta hai aur concrete ko kharab hone se bachata hai. Ye ek chemical process ko robotic process bana deta hai.

Market/Industrial Worth (Market mein ahmiyat)

- Target: B2B commercial construction aur civil engineering companies.
- Faida: Curing time ki guarantee deta hai, karoro ka nuksan bachata hai, aur heaters ki energy cost kam karta hai.

Prototype Scope (8-12 Months)

Ek chota concrete ka block banayein. Saste temperature/moisture sensors, ek Raspberry Pi, heat lamps aur chote water pumps use karke ek chota scale model banayein. AI agent ko Python (RL ya logic controller) mein program karein.

Commercialization Potential

Aap iska software construction tech companies (jaise Procore) ko license kar sakte hain, ya general contractors ko "Smart Blanket Kit" bana kar bech sakte hain.

2. ChemDose: Gande Paani (Wastewater) ki Agentic Safai

The Problem (Masla)

Textile mills aur chemical factories paani saaf karne wale chemicals par har saal bohot paisa zaya karti hain. Paani ki chemistry har ghante badalti rehti hai, isliye factory managers hamesha zaroorat se zyada chemical daal dete hain taake un par fine na lag jaye.

How It Works (Ye kaam kaise karta hai)

IoT chemical sensors gande paani ki tanki mein lagaye jate hain. Ek AI agent (Reinforcement Learning) paani ki badalti hui chemistry ko samajhta hai, aur pump ko control karke sirf aur sirf utne hi micro-drops chemical ke daalta hai jitni zaroorat hoti hai.

Why It's Novel (Ye naya kyun hai)

Water monitoring ek bohot aam FYP hai. Lekin AI ke zariye complex chemical reactions ko real-time mein control karna ek advanced industrial challenge hai jo aapki AI aur process control ki deep knowledge show karega.

Market/Industrial Worth (Market mein ahmiyat)

- Target: B2B manufacturing, textile mills, aur municipal water treatment plants.
- Faida: Chemical ki cost mein 20-40% kami aati hai, aur environmental compliance ki guarantee milti hai.

Prototype Scope (8-12 Months)

Ek benchtop water system banayein. Arduino ke sath saste pH sensors lagayein. Paani mein manually thora acid ya base milayein, aur AI agent ke zariye chhote pumps ko control karein jo paani ka pH level wapas normal kar dein.

Commercialization Potential

Factories ko apna poora plant change karne ki zaroorat nahi. Ye ek add-on module ki tarah unke mojudaa system mein fit kiya ja sakta hai.

3. TriageNode: Khud-kar Bio-Logistics Bachhao Agent

The Problem (Masla)

Jab life-saving vaccines ya organs le jane wale truck ka AC (cooler) raste mein kharab ho jata hai, toh saara saman waste ho jata hai. Insaan itni jaldi response nahi de pata ke koi aur rasta nikal sake ya naya truck bula sake.

How It Works (Ye kaam kaise karta hai)

Har keemti box mein ek IoT gateway laga hota hai. Agar temperature girne lagay, toh AI agent "jaag" jata hai, courier APIs (jaise Uber Freight, FedEx) check karta hai, qareebi cold-storage dhoondta hai, aur khud hi emergency rescue courier book karke payment bhi kar deta hai.

Why It's Novel (Ye naya kyun hai)

Isme AI khud ek mukhtar (independent) agent ban jata hai. Package ab sirf track nahi ho raha, balkay wo khud ek economic agent hai jo apni "jaan" bachane ke liye khud emergency qadam utha raha hai.

Market/Industrial Worth (Market mein ahmiyat)

- Target: Pharma logistics, organ transplant networks.
- Faida: Billions of dollars ka sensitive saman zaya hone se bachata hai.

Prototype Scope (8-12 Months)

Ek "smart cooler" banayein jisme Raspberry Pi, sim module, aur temperature sensors hon. Ek mock (fake) API environment banayein jahan AI agent failure detect karne ke baad successfully ek emergency ride "book" kar ke dispatch confirmation dikhaye.

Commercialization Potential

Is agentic software ko bari packaging aur cold-chain logistics companies ko license kiya ja sakta hai jo apne premium boxes mein ise integrate kar sakti hain.

4. VitiBot: Fasal ko Kooray (Frost) se Bachane Wala Agentic Swarm

The Problem (Masla)

Baaghat (jaise angoor ya seb) mein achanak thand (micro-frost) parne se fasal tabah ho jati hai. Kisaan bachne ke liye poore khet mein bohot bade heaters ya pankhey (wind machines) chala dete hain, jo bohot mehnga aur inefficient tareeqa hai.

How It Works (Ye kaam kaise karta hai)

Khet mein saste IoT weather sensors ka ek jaal (swarm) bicha hota hai. AI agent thandi hawa ke rukh aur "frost pockets" ko real-time mein map karta hai aur sirf unhi makhsoos hisson ke heaters/fans on karta hai jahan thand aayi hoti hai. Jaise jaise thand aage barhti hai, AI apne actions adjust karta hai.

Why It's Novel (Ye naya kyun hai)

Ye swarm intelligence aur dynamic physical actuation ka behtareen istemal hai. Ye sirf data nahi batata, balkay resources (fuel/energy) ko smart tarike se bacha kar faislay karta hai.

Market/Industrial Worth (Market mein ahmiyat)

- Target: Commercial viticulture aur high-value fruit orchards (baaghat).
- Faida: Fasal bhi bach jati hai aur poore khet mein heaters chalanay ki fuel cost bhi drastically kam ho jati hai.

Prototype Scope (8-12 Months)

Ek table par mitti aur 10-15 temperature sensors lagayein. Chote PC fans aur LEDs se heaters/fans ko represent karein. Dry ice ka istemal karke thand layein aur dikhayein ke AI sirf zaroorat ki jagah (path of the cold front) fans on karta hai.

Commercialization Potential

Agri-tech companies aur barey zameendaron ko iska software aur hardware system as a solution becha ja sakta hai.

5. ForgeAgent: 3D Printing mein Khud-kar G-Code Healing

The Problem (Masla)

Industrial 3D printing mein kai din lagte hain. Agar 40 ghante baad thori si bhi physical kharabi (jaise layer ka kharab hona) aaye, toh machine print karti rehti hai. Is se waqt, machine aur hazaron dollars ka material zaya ho jata hai. Mojooda smart printers sirf error detect karke print rok dete hain.

How It Works (Ye kaam kaise karta hai)

IoT acoustic sensors (mic) aur camera 3D print ko monitor karte hain. Jaise hi AI agent ko koi kharabi nazar aati hai, wo print rokne ke bajaye on-the-fly 3D printer ka G-code rewrite karta hai (jaise speed, heat, ya nozzle ki height badha dena) taake wo fault chalte chalte khud theek (heal) ho jaye.

Why It's Novel (Ye naya kyun hai)

Real-time mein chalte hue hardware ki manufacturing instructions ko change karna bohot mushkil kaam hai. AI ko physical mechanics theek karne ki authority dena software aur hardware engineering ka bohot impressive combination hai.

Market/Industrial Worth (Market mein ahmiyat)

- Target: Aerospace, automotive, aur high-end 3D printing farms.
- Faida: Machine ka time bachata hai, mehngi materials ko waste hone se roktah hai, aur 24/7 insaan ke dekh-bhaal ki zaroorat khatam karta hai.

Prototype Scope (8-12 Months)

Ek aam FDM 3D printer (jaise Ender 3) par webcam aur mic lagayein. Python-based agent banayein jo OctoPrint API se connected ho. Phir jaan boojh kar print mein error dalein (jaise filament ko thora rokna), aur dikhayein ke agent ne detect karke kaise G-code send kar ke nozzle ka temperature barha kar print theek kar liya.

Commercialization Potential

Industrial printer bananey wali companies ko ye as an enterprise software plugin (add-on) becha ja sakta hai.

1. ChromaAgent — Textile Dyehouse Controller

Simple Words Mein: Kapray rangne (textile dyeing) ke process ko autonomous (khudkar) banana.

- Problem: Kapra rangne ke liye master banda manually temperature, chemicals, aur pH set karta hai. Is tarah 8–15% kapra kharab (reject) ho jata hai, jis se paani, chemicals, aur bijli zaya hoti hai. Maujooda ERP ya SCADA systems sirf data save karte hain, khud se kuch theek nahi karte.
- How it Works: IoT sensors (pH, temperature, turbidity) lagaye jayein gay jo real-time data edge computer ko dein gay. Ek AI (Reinforcement Learning - RL) agent running batch ke dauran khud hi chemicals ki speed aur temperature ko sahi kare ga.
- Novelty (Nayi cheez): Aaj tak koi aisa system nahi hai jo har batch se seekh kar aglay batch ko behtar banaye (cross-batch memory).
- Market Value: Faisalabad aur Bangladesh ke textile mills is ke baray buyers hain. Agar 40% rejects bhi kam ho jayein, toh millions of rupees ki bachat hogi.
- Prototype (FYP Scope): Ek single dye vessel (tanki) par 6-8 sensors lagao. Pehle simulation par AI train karo, phir real factory mein test karo.

2. GrainMind — Post-Harvest Storage Intelligence

Simple Words Mein: Godowns (silos) mein padi gandum aur chawal ko kharab hone se bachana.

- Problem: Pakistan aur South Asia mein 14–22% gandum aur chawal storage ke dauran moisture (nami) aur keeron ki wajah se kharab ho jata hai. Abhi sirf alerts aate hain, par jab tak banda action leta hai, der ho chuki hoti hai.
- How it Works: Godown ke andar temperature cables, humidity sensors, aur CO_2 sensors (jo keeron ke saans lene ko detect karte hain) lagayein gay. AI Agent khud hi aeration fans on kare ga ya fumigation (spraying) start kar de ga.
- Novelty: Pehle wale systems sirf data dikhate hain (passive). Ye AI agent khud decision le ga aur action lay ga (active network management).
- Market Value: PASSCO, Punjab Food Department, aur baday flour mills iske customer hain. Thodi si bachat se bhi karoroon dollar ka faida hoga.
- Prototype (FYP Scope): Kisi local flour mill ke 10–20 bins par Raspberry Pi aur saste sensors (DHT22, DS18B20) ke sath LoRaWAN lagao aur system design karo.

3. VentIQ — Underground Mine Ventilation Agent

Simple Words Mein: Underground kaan (mines) mein taaza hawa (ventilation) ka kharcha kam karna aur safety barhana.

- Problem: Underground mines mein 40–60% bijli sirf baray baray fans chalane mein zaya hoti hai. Fans har waqt full speed par chaltay hain, chahe wahan mazdoor ho ya na ho. Is ke ilawa zehrili gases (Methane, Carbon Monoxide) ka khatra rehta hai.
- How it Works: Mazdooron ko BLE/UWB location tags diye jayein gay aur pooray tunnel mein gas sensors lagayein gay. AI Agent ko mine ka map (graph) pata hoga. Wo fans ki speed aur doors ki position ko khud control kare ga jahan zaroorat hogi.
- Novelty: Ye sirf rule-based nahi hoga balkay dynamic hoga. AI pehle se hi guess kar le ga ke kis area mein hawa ki zaroorat hone wali hai.
- Market Value: Coal aur metal mines isko khareedein gi. Ek mid-size mine ka saal ka \$750K–\$1.5M tak ka bijli ka bill bach sakta hai.
- Prototype (FYP Scope): PVC pipes se 1:20 scale ka ek chota mine tunnel network (testbed) banao. Chote fans aur sensors laga kar AI (DQN/PPO) ko train karo.

4. AquaAgent — Controller for Intensive Aquaculture

Simple Words Mein: Machli aur jhenga (shrimp) ki farming ke talabon (ponds) ko automated control karna.

- Problem: Intensive fish farming mein raat ke waqt oxygen (\$DO\$) achanak crash kar jati hai, jis se 15–25% machliyan mar jati hain. Aerators (oxygen chalane wale pankhe) ko 24/7 chalane se bijli ka kharcha bohot aata hai.
- How it Works: IoT nodes har pond ka Dissolved Oxygen, pH, aur temperature check karein gay. Ek central AI agent pooray farm ke aerators aur automatic feeding machines ko control kare ga.
- Novelty: Agar pooray farm mein bijli ki limit kam ho, toh AI khud decide kare ga ke kis talab ko pehle oxygen deni hai (Constrained multi-pond resource allocation).
- Market Value: Thailand, Vietnam, aur India ke shrimp farmers is ke customer hain. Ye machliyon ko marne se bachaye ga aur bijli ka bill 60% tak optimize kare ga.
- Prototype (FYP Scope): 3-5 ponds ka data lekar simulation banao. Weather API connect karo taake badal aur dhoop ke mutabiq oxygen level control ho sakay.

5. SinterAgent — Kiln Controller for Technical Ceramics

Simple Words Mein: Ceramic (tiles, dental blanks, furnace bricks) banane wali bhatti (kiln) ke temperature aur gas ko control karna.

- Problem: High-temperature bhattiyon mein 8–20% ceramic items crack ya warp ho jati hain kyunke bhatti ke andar ka atmosphere (Oxygen aur CO/CO_2 ratio) manually handle hota hai jo bohot mushkil kaam hai.
- How it Works: Bhatti ke andar high-temperature thermocouples aur gas analyzers lagayein gay. AI Agent khud hi gas-to-air ratio, conveyor belt ki speed, aur nitrogen injection control kare ga.
- Novelty: Temperature, gas chemistry, aur speed teeno ko ek sath (joint policy) manage karne wala koi system market mein nahi hai.
- Market Value: Advanced ceramics market bohot badi hai (\$12B+). Bhatti ka defect rate aadha karne se daily hazaron dollars ki bachat hoti hai.
- Prototype (FYP Scope): Ek choti lab furnace/kiln par sensors lagao. Sintering kinetics ki equations use kar ke physics-informed AI model train karo.

Quick Comparison Table

#	Title	Main Tech (RL Action Space)	Kis Ko Faida Hoga?
1	ChromaAgent	Chemical dosing & Temp management	Textile Mills (Faisalabad, etc.)
2	GrainMind	Fan scheduling & Fumigation timing	Govt Grain Silos / PASSCO
3	VentIQ	Fan speed & Gate positions in tunnels	Mining Sector
4	AquaAgent	Aerator power allocation & Feeding rate	Fish/Shrimp Commercial Farms
5	SinterAgent	Air-gas ratio & Conveyor speed	Ceramic Manufacturing

💡 FYP Pro-Tip: In sab projects mein hardware pehle din se banana mushkil hota hai. Pehle 1-3 months sirf Simulation (Software) par AI train karo (using Python/OpenAI Gym). Jab software chal jaye, tab physical sensors aur hardware par kaam start karo. Is tarah tumhara project kabhi fail nahi hoga!

Yar, ye 5 fresh aur solid Final Year Project (FYP) ideas hain jo bilkul Pakistan ke industrial problems (bhatta, kapra, load-shedding, aata, aur aam/kinnu) ko hal karne ke liye design kiye gaye hain. Inko simple Roman Urdu aur English mix mein easily samajh sako gay:

01 · KilnMind — Brick Kilns (Bhatte) ka AI Controller

Simple Words Mein: Bhatte (Brick Kiln) ke andar koalay (coal) ka kharcha aur dhuwan kam karna.

- Problem: Pakistan mein takreeban 20,000 bhatte hain. Wahan koala manually dala jata hai aur dhuwan/temperatures ko labour apne andaze se control karti hai. Is wajah se 30-40% koala zaya hota hai aur pollution bohot phelti hai. Abhi tak is sector mein koi automation nahi hai.
- How it Works: Bhatte ki firing zone mein high-temperature J-type thermocouples (sensors) aur draft pressure sensors lagayein gay. Ek AI Agent (PPO Algorithm) data check kar ke khud hi dampers (hawa ke darwaze) aur fuel feed ko adjust kare ga taake har taraf barabar temperature rahe.
- Novelty (Nayi cheez): Bhatton ke liye aaj tak koi aisa automated intelligent system nahi bana jo live sensor data dekh kar koalay ki bachat khud kare.
- Pakistan Fact & Worth: Punjab EPA ke rules aur koalay ki barhti qeemat ki wajah se bhatta maalik pareshan hain. Agar 10% koala bhi bache, toh ek bhatte ka roz ka 30,000 se 75,000 Rupee bache ga.
- Prototype (FYP Scope): Sheikhpura ya Lahore ke kisi bhatte par 8 thermocouples aur Raspberry Pi laga kar system test karo. Pehle Python (Stable Baselines3) par simulator banao.

02 · DyeAgent — Textile Dyehouse Filter & Dosing

Simple Words Mein: Faisalabad ke textile mills mein kapray ka rang (dyeing) pehli baar mein bilkul perfect nikalna.

- Problem: Pakistan ke dyehouses mein "Right-First-Time" (RFT) rate sirf 60-70% hai (yani 100 mein se 30-40 batch kharab hotay hain aur unko dobara rangna parta hai). Is se paani, chemical, aur gas zaya hoti hai. Turkish mills ka rate 90% tak hota hai.
- How it Works: Dyeing tank ke andar fiber-optic spectrophotometer (jo rang check karta hai) aur pH/conductivity sensors lagao. AI Agent (SAC Algorithm) live rang aur pH dekh kar automatic dosing pumps ko micro-commands bheje ga ke kab kitna chemical aur color mazeed dalna hai.
- Novelty: Maujooda software sirf purana recipe data dikhate hain. Ye AI system chalti hui machine ke andar live color sahi kare ga.
- Pakistan Fact & Worth: Faisalabad mein 400+ dyehouses hain. Agar RFT rate behtar ho jaye toh ek factory ka mahine ka 20 se 40 Lakh Rupee bache ga. Saath hi EU ke ZDHC environmental standards bhi poore hon gay.
- Prototype (FYP Scope): Ek 5-liter ka chota lab-scale dyeing setup banao. Peristaltic pumps aur saste Arduino/Raspberry Pi se dosing control karo. Phir Faisalabad ki kisi mill ke live data par validate karo.

03 · GridShift — Load-Shedding-Aware Factory Scheduler

Simple Words Mein: Factory ki production ko load-shedding ke schedule ke mutabiq chalana taake mehnga diesel generator kam se kam chale.

- Problem: Garmiyon mein factories ko 8-14 ghante load-shedding ka saamna karna parta hai. Wapda/K-Electric ki bijli Rs. 20-28 hai jabki diesel generator Rs. 35-50/kWh parta hai. Factories achanak light janay par react karti hain, pehle se planning nahi karti.
- How it Works: Factory ki har line par smart IoT sub-meters lagao. AI Agent (DQN Algorithm) factory ke DISCO feeder ki pichli tareekh (history) se seekh jaye ga ke kab light janay ka kitna chance hai. Wo 4-8 ghante pehle hi heavy machines ka schedule set kar de ga, battery/UPS charge kar le ga, aur non-critical kaam sasti bijli ke time par shift kar de ga.
- Novelty: Duniya ke baqi countries mein outage survival par kaam nahi hota (wahan light nahi jati). Ye software khaas tor par Pakistan ke grid system (LESCO, FESCO, MEPCO) ke liye moated solution hai.
- Pakistan Fact & Worth: Pakistan mein 9,000+ baday manufacturing units hain. Sirf 20% generator ke ghante kam karne se SMEs ka mahine ka 1 se 2 Lakh Rupee bache ga. LCCI aur FCCI ke zariye isko becha ja sakta hai.
- Prototype (FYP Scope): Shelly EM ya saste CT-clamp meters factory lines par lagao. LESCO/FESCO se 2 saal ka outage data nikal kar Python/OpenAI Gym mein factory environment banao aur test karo.

04 · GrainMind — Wheat Blending Optimizer for Flour Mills

Simple Words Mein: Flour mills (aate ki mill) mein mukhtalif qism ki gandum ko is tarah mix karna ke saste mein behtareen aata (chapati/bread flour) banay.

- Problem: Pakistan mein 1,100+ flour mills hain. Har gandum ka protein, moisture (nami), aur ash (raakh) level alag hota hai. Abhi lab test ki report 2-4 ghante mein aati hai, tab tak andaze se gandum mix kar di jati hai. Is se ya toh bohot mehangi gandum zaya hoti hai ya aate ki quality kharab ho jati hai.
- How it Works: Gandum uthane wali conveyor belts par NIR (Near-Infrared) sensors lagayein gay jo real-time mein protein aur moisture batayein gay. AI Agent (TD3 Algorithm) blending hoppers ke gates (actuators) ko khud hi kam ya zyada khole ga taake sasti gandum zyada use ho aur standard poora rahay.
- Novelty: European systems (Bühler wagara) bohot mehenge hain (\$500K+). Saste NIR sensor ke sath cost-minimizing RL agent lagana bilkul naya concept hai.
- Pakistan Fact & Worth: Millers Association (PFMA) ke mutabiq sasti blending se rozana Rs. 50,000 se 450,000 tak ki bachat ho sakti hai ek mill ke liye. Afghanistan aur Gulf export ke liye quality verification aasan ho jaye gi.
- Prototype (FYP Scope): Ek chota 3-bin blending rig model banao. Hamamatsu C12666MA jaisa chota NIR micro-spectrometer aur servo motors use karo. 3-5 Pakistani gandum ki varieties (Inqlab-91, Seher, etc.) ka data use karo.

05 · ColdAgent — Cold Chain Orchestrator for Mango & Citrus Export

Simple Words Mein: Aam (Mangoes) aur Kinnu ko farm se lekar port tak kharab hone se bachana taake export barh sakay.

- Problem: Pakistan mein 35-40% aam aur kinnu cold chain kharab hone ki wajah se raste mein hi gal sarr jatay hain. Refrigerated containers (reefers) toh hain, par unko raste mein koi autonomously track aur target change nahi karta. Ethylene gas barhne se aam jaldi pak jata hai.
- How it Works: Packhouses, cold rooms, aur reefer containers mein IoT nodes (4G trackers + Ethylene sensors + \$CO_2\$ + Temp) lagayein gay. Ek multi-agent AI system (MAPPO Algorithm) sab nodes ko aik sath sambhale ga. Agar kisi container mein aam jaldi pak raha ho, toh AI khud hi reefer ka temperature setpoint remote control se kam kar de ga ya truck ka rasta change karwa de ga.
- Novelty: Abhi tak sirf alerts aate hain dashboard par (passive tracking). Ye pehla system hai jo multi-agent RL ke zariye pre-cooling, storage, aur transit ko khud hi micro-manage kare ga.
- Pakistan Fact & Worth: TDAP (Trade Development Authority of Pakistan) ke mutabiq agar cold chain sahi ho, toh mango export \$100M se \$300M+ tak ja sakti hai. Multan, Rahim Yar Khan, aur Sargodha ke export zones is ke direct buyers hain.
- Prototype (FYP Scope): ESP32, DS18B20 temperature probes, MQ-3 sensor (for ethylene proxy), aur ek GPRS module use kar ke hardware banao. Mango season (May-August) mein kisi cold store par test karo.

Comparison Summary Table

#	Project Name	AI Algorithm	IoT Sensors Layer	Pakistan Edge / Benefit
01	KilnMind	PPO	Thermocouples & Draft Sensors	20,000 Bhatte hain, jahan zero automation hai.
02	DyeAgent	SAC	Spectrophotometer & pH Probes	Low RFT rate (60-70%) ko global level (90%) par lana.
03	GridShift	DQN	Smart Sub-Meters (Shelly/CT Clamps)	Pakistan ke unique load-shedding load curves par customized.
04	GrainMind	TD3	Mini NIR Spectrometer & Load Cells	1,100 Flour mills ke liye sasta automated blending tool.
05	ColdAgent	MAPPO	Temp/Humidity/Ethylene + GPS (4G)	\$300M mango/citrus export potential ko unlock karna.

💡 FYP Success Formula: In paanchon projects mein Hardware Deployment sab se badha risk hota hai (mill owners jaldi apni chalte plant par testing nahi karne dete). Iska hal ye hai ke pehle 3-4 months sirf Python mein Simulation environment banao. Jub simulation chal jaye aur tumhare paas graphs aa jayein, phir hardware par haath daalo. Is se project backup par rehta hai aur external delays se FYP fail nahi hota!